Exploratory Data Analysis
Online Retail Sales

Transactions from 2010-12-01 to 2011-12-09

EDA Capstone Project

Data notice: this report was generated from a **synthetic sample** that mirrors the structure of the Online Retail dataset. The figures below are illustrative only. Replace *Dataset/Online_Retail.xlsx* with the real file and re-run the notebook to produce real results.

1. Introduction and objectives

This project explores transaction data from a UK-based online retailer that sells giftware, much of it to wholesale customers. The objectives are to (1) assess data quality and prepare the data for analysis, (2) describe how sales change over time, (3) identify the products that drive revenue and (4) understand which countries contribute most.

2. Dataset overview

The raw file holds **48,694** line items across eight columns (InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country). Each row is one product line on an invoice.

3. Data cleaning

Cancellations made up 1.9% of lines and 24.7% of rows had no CustomerID. Rows without a CustomerID were kept for sales analysis but excluded from customer-level metrics.

Cleaning step	Rows removed
Remove duplicate rows	250
Remove cancelled invoices	901
Remove non-positive quantity/price	60
Remove missing descriptions	78
Remove non-product codes	39
Rows retained	47,366

4. Headline metrics

Metric	Value
Total revenue	GBP 1,066,119
Completed orders	5,196
Identified customers	2,046
Distinct products	320
Countries	14
Average order value	GBP 205
Median order value	GBP 108

5. Findings

5.1 Sales over time

Revenue peaks in **2011-11** at GBP 162,091, while the weakest complete month is **2011-04**. The final month is incomplete, so its total understates a full month. Orders are concentrated on **Wednesday** and around **12:00**.

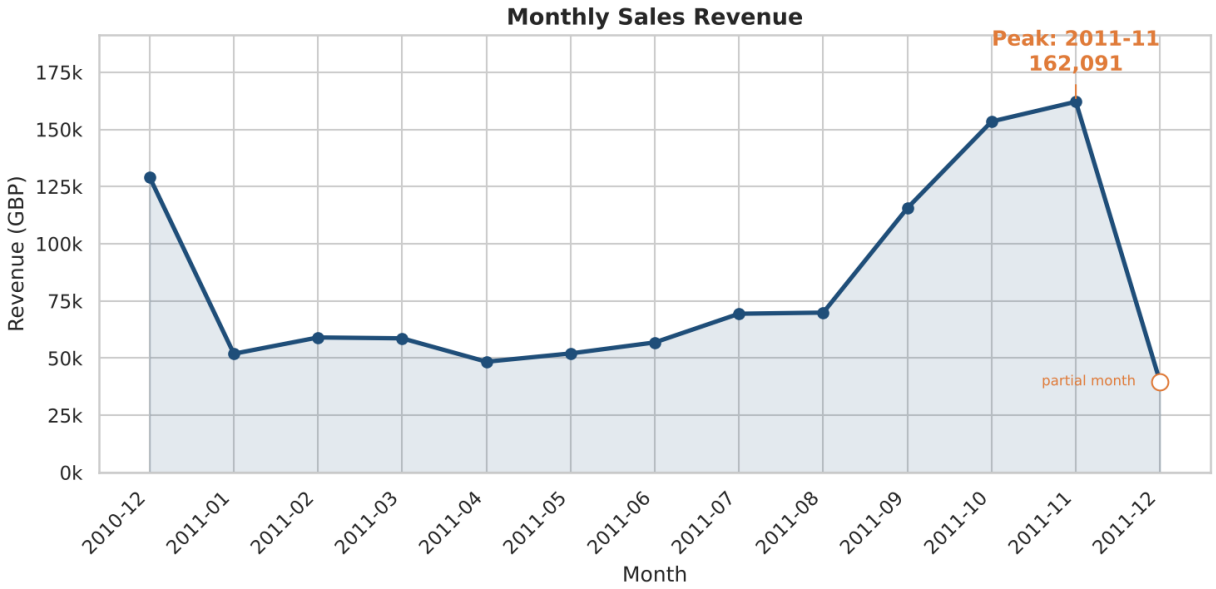


Figure 1. Monthly sales revenue

5.2 Top products

The best-selling product by revenue is **WHITE STORAGE CUSHION** (GBP 64,076). The top 10 products together account for 30.6% of revenue.

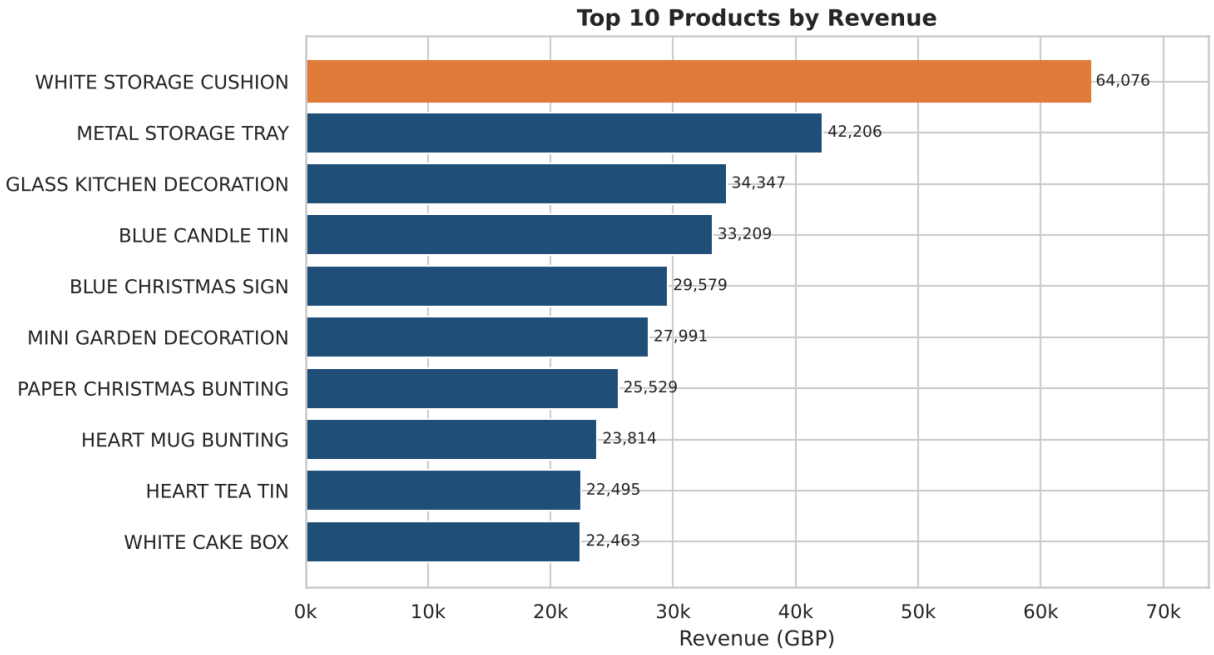


Figure 2. Top 10 products by revenue

5.3 Sales by country

United Kingdom generates 87.0% of revenue. Excluding it, the largest market is **Germany** (GBP 30,000). Among identified customers, the top 20% account for 59.9% of their revenue.

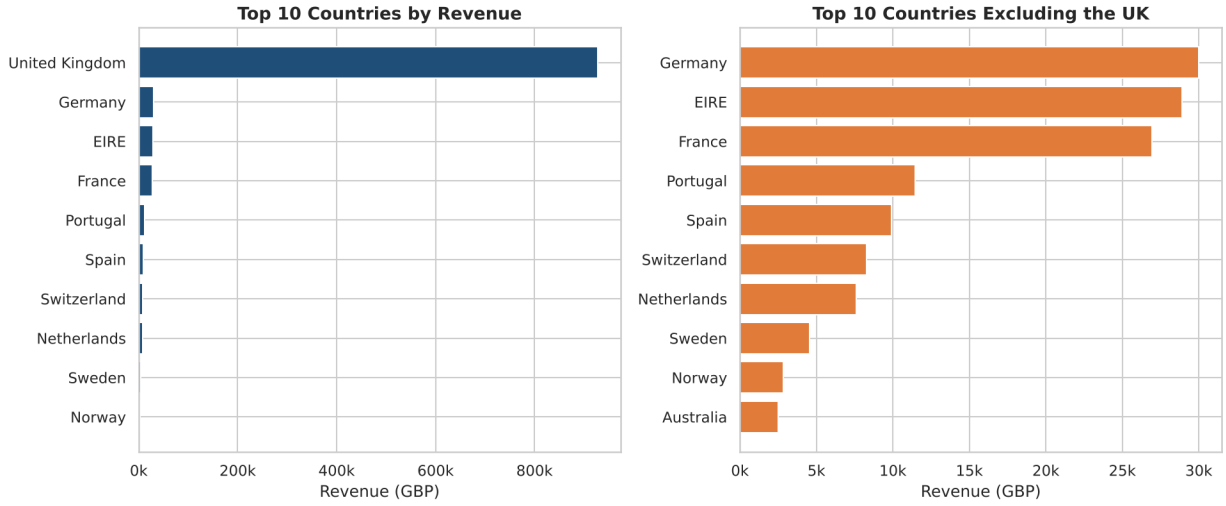


Figure 3. Revenue by country, with and without the UK

6. Recommendations

• **Plan inventory around the seasonal peak**, building stock ahead of the highest-revenue months.
• **Protect best sellers** by monitoring stock levels on the top products, which carry a large share of revenue.
• **Develop international markets**, starting with the largest non-domestic country, since revenue is heavily domestic.

• **Improve customer tracking**: missing CustomerIDs limit retention and segmentation analysis.
• **Investigate cancellations** to find product or fulfilment issues behind returned orders.

7. Limitations

The data covers about one year, so seasonality cannot be separated from growth with confidence. Cancelled orders were removed rather than netted off, so revenue is gross of returns, and rows without a CustomerID cannot be linked to individual customers. This analysis is descriptive and does not establish causes.